A Unified AI and Blockchain Framework for Jasmine Disease Detection, Yield Prediction, Smart Irrigation, Harvest Planning, and Traceability

Industry Project : 2026-27

Name of the Industry :
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Introduction to the Project

- Jasmine cultivation faces challenges like disease detection delays and yield loss.
- This project develops an AI-based Smart Monitoring System for early disease detection and yield prediction.
- Provides real-time, data-driven insights to improve productivity and promote sustainable farming.



Problem Statement

- Manual disease detection is time-consuming and inaccurate, relying heavily on farmer expertise.
- Late diagnosis reduces crop quality, yield, and profitability.
- Lack of integrated systems combining disease detection, environmental monitoring, and advisory support.
- No real-time, crop-specific recommendations for jasmine plants.
- Incorrect treatments increase costs and environmental impact.
- Need for an AI-based automated system for early detection, yield prediction, and data-driven decision support.



Lacuna in the Existing System

EXISTING SOLUTIONS

Plantix

Features:

1. AI-based plant disease detection from mobile
2. Treatment recommendations
3. Multi-language support
4. Community & expert Q&A

Gaps:

1. Not jasmine-specific
2. Cannot detect early bud/flower issues
3. No severity scoring
4. Generic treatment advice

LeafDoctor

Features:

1. Measures leaf disease area (%)
2. Useful for academic severity analysis
3. Simple and lightweight tool

Gaps:

1. Manual process
2. Cannot identify disease type automatically
3. Leaf-only analysis
4. No treatment suggestions or recommendations

Agrio

Features:

1. AI-based plant disease detection
2. Provides early disease alerts based on weather conditions
3. Field scouting & reporting tools

Gaps:

1. Focused on food crops
2. Weather-heavy, weak visual accuracy for jasmine
3. No visual disease explanation
4. Not localized for Indian jasmine farmers

PlantIn

Features:

1. Basic plant disease detection
2. General plant care guidance
3. User-friendly, beginner-oriented app

Gaps:

1. Not designed for commercial crops
2. Detects stress, not disease
3. No progression tracking
4. No farm-level integration
5. No weather integration

Literature Survey

Author & Year	Title	Method Used	Findings	Limitations
KVK Erode (2022–2025)	Impact of timely application of fertilisers on yield and economics of Jasmine (Jasminum sambac L.)	Field comparison of fertilizer practices	Timely balanced fertilization increased yield and profit	Region-specific study
Sharma et al. (2022)	Innovative Design of an Experimental Jasmine Flower Automated Picker System Using Vertical	YOLOv5 detection + robotic gripper	Accurate bud detection and automated picking	Prototype stage
Karthikeyan P et al.	An Attention-based Deep Learning Model for Jasminum Quality Grading	InceptionResNetV2 with attention	~92% grading accuracy	Small dataset
Rao & Iyer (2021)	A Machine Learning Approach for Predictive Analysis of Jasmine Flower Yield and Plant Health	Random Forest + CNN using sensor & image data	Predicts yield and plant health	Needs sensors & large data
Gupta et al. (2023	InsightNet: A Deep Learning Framework for Enhanced Plant Disease Detection and Explainable Insights	Improved MobileNet + Grad-CAM	~98% disease detection accuracy	High computation

Datasets

Arabian Jasmine Leaf Condition Dataset (AJLCD-2025)

Published: 21 July 2025 | Version 1 | DOI: 10.17632/nwdzwk89zs.1
Contributors: Rokonozzaman Ayon, Nur Yea Afroz Suchi, Md Asif Shahriar Arpon, Tanjina Ahmed Tuly Tanjina Ahmed Tuly, Abdullah Al Noman

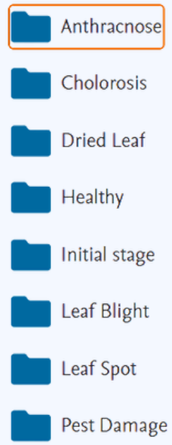
Description

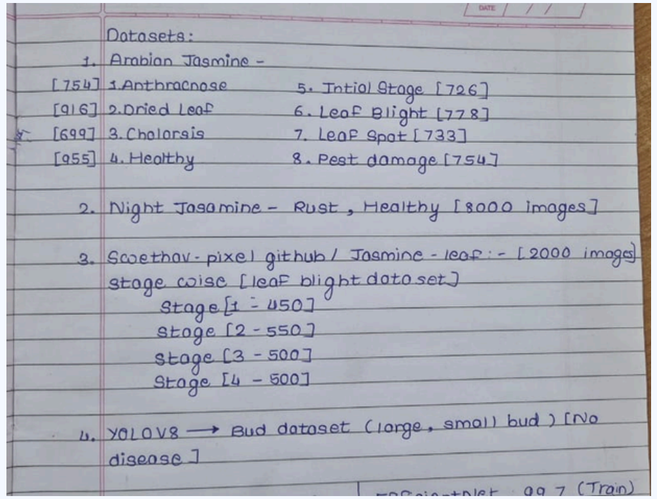
The Arabian Jasmine Leaf Condition Dataset (AJLCD-2025) consists of 6,315 high-resolution images of Arabian Jasmine (Jasminum sambac) leaves, collected in 2025 from Sonargaon, Narayanganj, Bangladesh (GPS: 23.6303°N, 90.6028°E; altitude: 45.2 meters). The original images were captured using a Xiaomi M2102J20SG smartphone under natural daylight conditions, and contain full EXIF metadata such as F-stop (f/1.8), ISO speed (ISO-96), exposure time (1/120 sec), focal length (5 mm), metering mode (center-weighted average), and white balance set to auto. The raw images are in JPG format, primarily with dimensions of 6000×8000 pixels (some 4000×3000), 24-bit sRGB color space, and 72 dpi resolution.

To prepare the dataset for deep learning and image classification tasks, all images were preprocessed by resizing them uniformly to 1440×1080 pixels, applying background removal to isolate leaf structures, and replacing the background with a clean white canvas to enhance clarity and model performance. The dataset includes 8 different types of leaf conditions, representing various disease conditions and stages of the Arabian Jasmine plant: Anthracnose (754 images), Chlorosis (699), Dried Leaf (916), Healthy (955), Initial Stage (726), Leaf Blight (778), Leaf Spot (733), and Pest Damage (754).

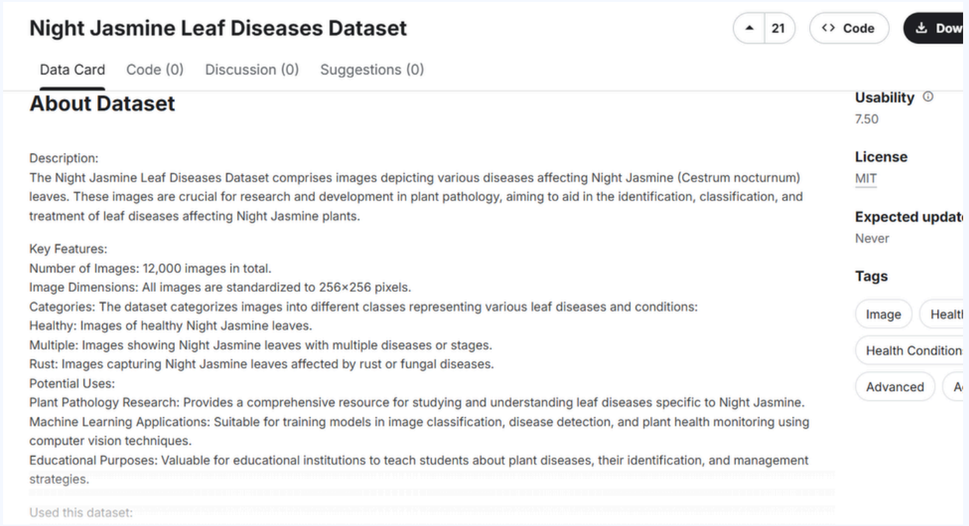
This AJLCD-2025 dataset is built to support many applications like leaf disease classification using CNNs and transformer models, mobile app development for plant disease diagnosis, explainable AI (XAI) research, and smart farming solutions. It is also highly relevant for fields like Agrotechnology, where it can aid in developing advanced crop management strategies, improving plant health monitoring, and optimizing sustainable agricultural practices. The hope is that this dataset will push forward progress in precision agriculture, agro-tech innovation, and AI-driven plant pathology.

Dataset is divided into this classes :

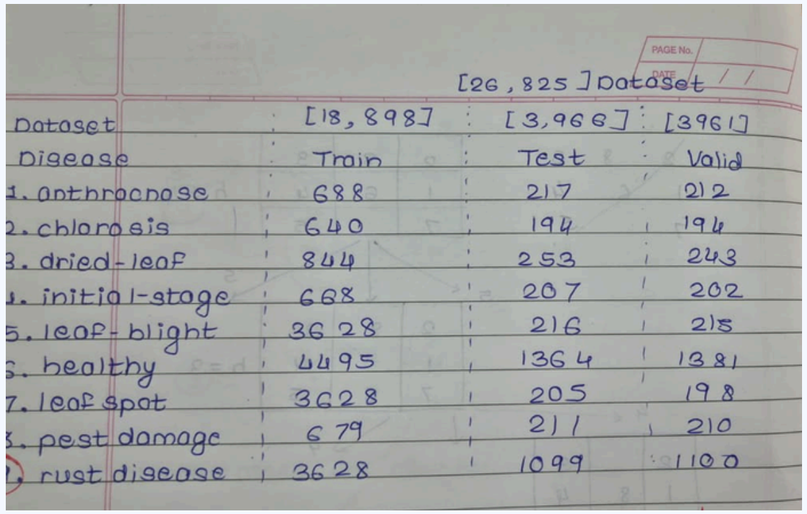




Night Jasmine Leaf Diseases Dataset



Combined Dataset



Problem Statement

- **Market demand and prices fluctuate depending on festivals, weather, flower quality, and local market conditions.**
- **Harvesting too early or producing more flowers than buyers require can lead to financial losses.**

Crop information	Jasmine variety, farm area, previous yield
Flower quality	Flower quality, disease severity
Market information	Historical market price, previous demand, market location
Festival information	Festival/event indicator, festival intensity
Weather	Weather conditions
Harvest readiness	Days to full bloom
Logistics	Transportation distance
Buyer information	Buyer capacity
Calendar information	Date-related features

Our Solution

We developed an AI-based prediction module that estimates:

- **Expected local market demand.**
- **Expected jasmine price range.**
- **Quantity likely to be accepted by buyers.**
- **Risk of unsold or overripe flowers.**
- **A suitable harvesting date based on market conditions.**

Dataset

- **Dataset size: 10,000 records (Synthetic)**
- **Split ratio: 80:20**

Models used

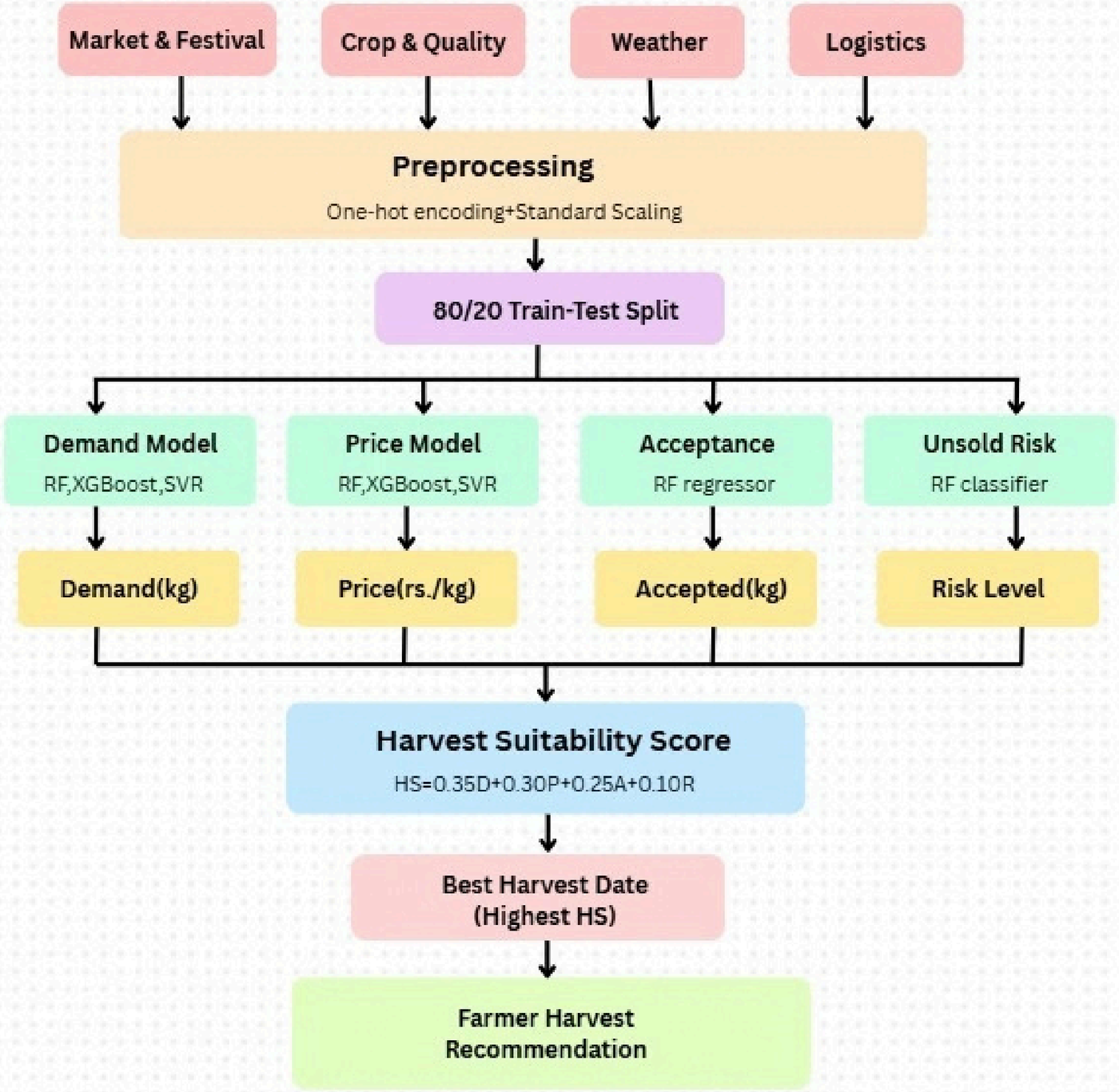
- **Random Forest**
- **XGBoost**
- **SVR**

Market demand:

- **The XGBoost demand model achieved an R^2 score of 0.850**

Unsold risk:

- **The Random Forest classifier achieved 55.05% accuracy,**


Market & Festival
Crop & Quality
Weather
Logistics
Preprocessing
One-hot encoding+Standard Scaling
80/20 Train-Test Split
Demand Model
RF,XGBoost,SVR
Price Model
RF,XGBoost,SVR
Acceptance
RF regressor
Unsold Risk
RF classifier
Demand(kg)
Price(rs./kg)
Accepted(kg)
Risk Level
Harvest Suitability Score
HS=0.35D+0.30P+0.25A+0.10R
Best Harvest Date
(Highest HS)
Farmer Harvest
Recommendation

Smart Irrigation Advisor for Jasmine Cultivation

AI-driven irrigation decision and water quantity recommendation for sustainable jasmine farming

1 Objective & Problem



Traditional irrigation is often based on fixed schedules or experience, leading to overwatering or underwatering.

Our goal is to help jasmine farmers decide when irrigation is required and how much water should be supplied per plant using machine learning.

2 Dataset Overview

Dataset Name	jasmine_smart_irrigation_final_20000.csv
Number of Records	20,000
Total Columns	42
Data Type	Synthetic dataset (generated for model development)

soil_moisture	temperature	humidity	rainfall	growth_stage	...
32.5	28.1	65.2	0.0	Vegetative	...
41.8	30.5	72.1	5.2	Budding	...
18.3	27.9	58.4	0.0	Flowering	...

3 Input Parameters (Features)

Soil Parameters
- Soil moisture, soil type, soil temperature
- Field capacity, wilting point, root-zone depth

Weather Parameters
- Rainfall (last 24h, forecast 24h/48h)
- Temperature, humidity, wind speed
- Solar radiation

Crop Parameters
- Growth stage, days after pruning
- Crop coefficient (Kc), disease severity

Irrigation History
- Days since last irrigation
- Previous irrigation amount
- Irrigation method, water used in last 7 days

Water Requirement Parameters
- Reference evapotranspiration (ET0)
- Crop evapotranspiration (ETc)
- Effective rainfall, water deficit

4 Methodology / Workflow

Dataset
(20,000 records, 42 features)

Data Preprocessing
- Handle missing values
- Encode categorical features
- Scale numerical features

Classification Models (Irrigation Required)
- Random Forest
- XGBoost
- Logistic Regression
- SVM

Regression Models (Irrigation Quantity)
- Random Forest Regressor
- XGBoost Regressor
- SVR
- Gradient Boosting Regressor

Prediction Output
- Whether irrigation is required (Yes/No)
- Recommended water quantity (L/plant)

5 Model Performance

Classification Results (Irrigation Required)

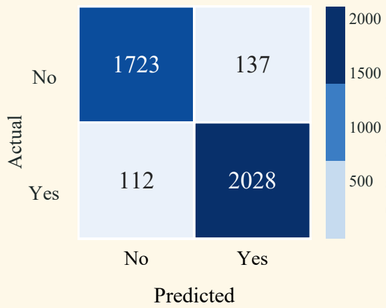
Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)
Logistic Regression	82.14	81.02	83.26	82.12	89.10
SVM	83.67	82.91	84.33	83.61	91.25
Random Forest	85.92	26.19	85.21	85.65	93.48
XGBoost	**87.50**	**87.21**	**87.63**	**87.42**	**94.95**

Regression Results (Irrigation Quantity in L/plant)

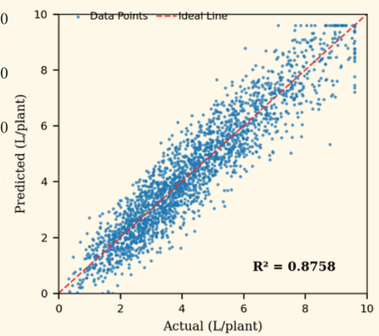
Model	MAE	RMSE	R^2	MAPE (%)
Random Forest	1.32	1.95	0.8123	6.24
SVR	1.18	1.76	0.8435	5.61
Gradient Boosting	1.05	1.62	0.8621	4.88
XGBoost	**0.89**	**1.43**	**0.8758**	**4.46**

6 Result Visualizations (XGBoost)

Confusion Matrix (Classification)



Actual vs Predicted Irrigation Quantity



7 Example Prediction (App Output)

Jasmine Smart Irrigation Advisor

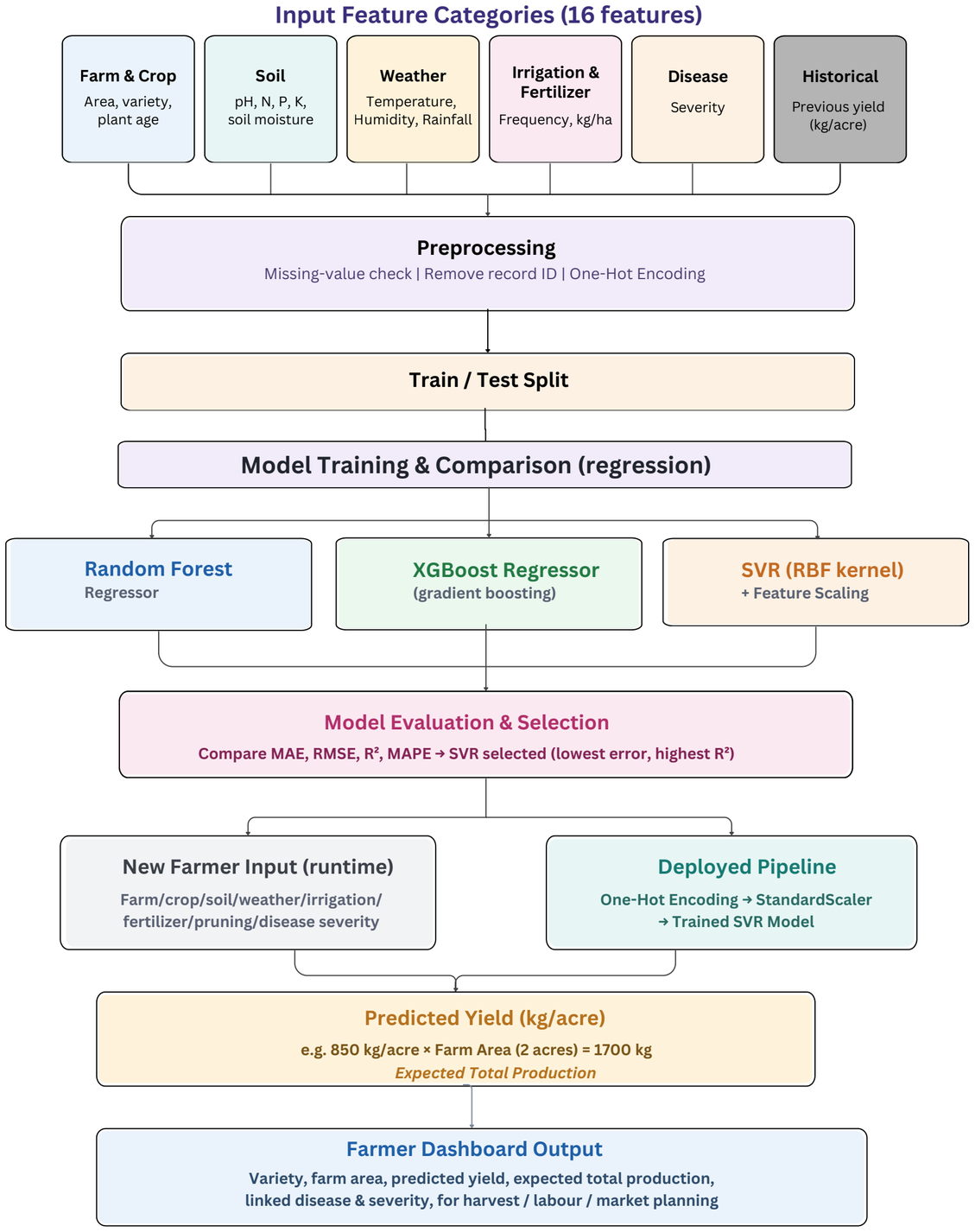
Soil Moisture	22.5 %
Temperature	29.1 °C
Humidity	64 %
Rainfall (24h)	0.0 mm
Growth Stage	Flowering
Days Since Last Irrigation	5 days

Irrigation Required: Yes
Recommended Water Quantity:
3.6 L / plant

8 Key Insights & Future Scope

- Helps farmers make informed irrigation decisions and reduces unnecessary water usage.
- XGBoost performed best with 87.50% classification accuracy and $R^2 = 0.8758$ for quantity prediction.
- Supports sustainable and efficient use of water resources.
- Future work: integrate real farm data, soil-moisture sensors and live weather APIs for real-time recommendations.

JASMINE YIELD PREDICTION



MODEL PERFORMANCE COMPARISON

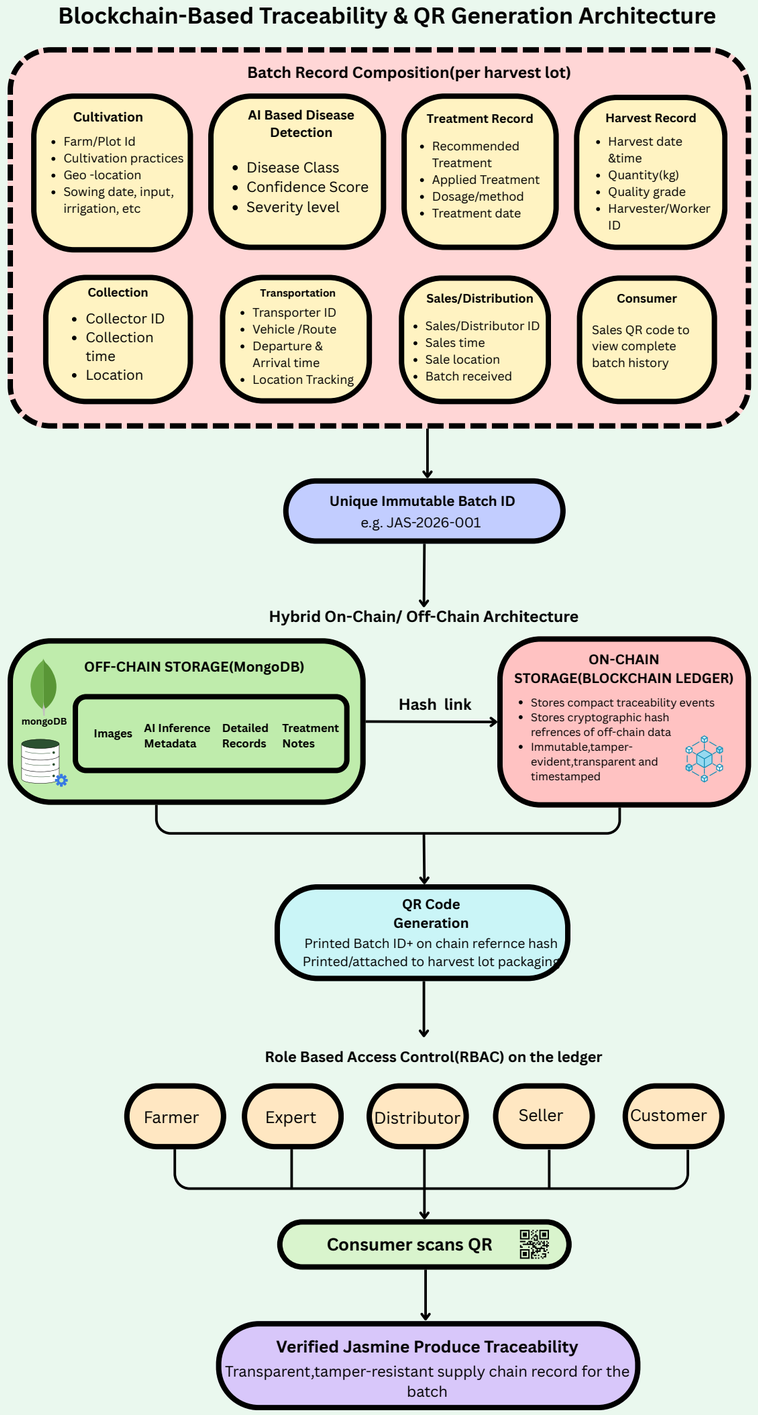
Model	MAE (kg/acre)	RMSE (kg/acre)	R^2 (Higher is better)	MAPE (%)
Random Forest	84.79	104.99	0.7472	8.37
XGBoost	68.94	86.63	0.8279	6.74
SVR (Selected)	**67.94**	**85.81**	**0.8311**	**6.63**

SVR achieved the best performance with highest R^2 and lowest error values.

EXAMPLE OUTPUT

Variety	Farm Area	Predicted Yield	Total Expected Yield
Mogra	2 acres	1299.05 kg/acre	2598.09 kg

Total Expected Yield = 1299.05 kg/acre × 2 acres = 2598.09 kg



Blockchain-Based Jasmine Produce Traceability and QR Code Generation Architecture

What the blockchain is storing

The blockchain is not storing the heavy data — no images, no full text records. It only stores small, structured entries for each event, like:

- **Batch ID** (e.g., JAS-2026-001)
- **Event type** (cultivation logged, disease detected, treatment applied, harvest recorded, collected, transported, sold)
- **Timestamp** of when the event happened
- **Actor/role** who recorded it (which farmer, which distributor, etc.)
- **A cryptographic hash (like SHA-256)** of the corresponding detailed record that actually lives in MongoDB

```
Batch ID: JAS-2026-001
Event: Disease Detection
Timestamp: 2026-06-14 09:32
Recorded by: Farmer_102
Hash: 4f2a9c...e7b1  (points to the full MongoDB record)
```

Methodology Employed

Data Acquisition
- Upload jasmine leaf images.

Image Preprocessing & Segmentation
- Image resizing
- Noise removal
- Contrast enhancement

Feature Extraction
- Automatic feature extraction via deep learning
- ~~Color,~~ → texture, and pattern recognition

Deep Learning Models
- EfficientNet-B0→ High accuracy (best performing model)

Classification (Softmax Layer)
- Converts outputs into probability scores
- Identifies most likely disease class

Disease Detection & Severity Analysis

Classifies:
- → Healthy / Diseased
- → Disease type (leaf spot, etc.)

Yield Prediction Module

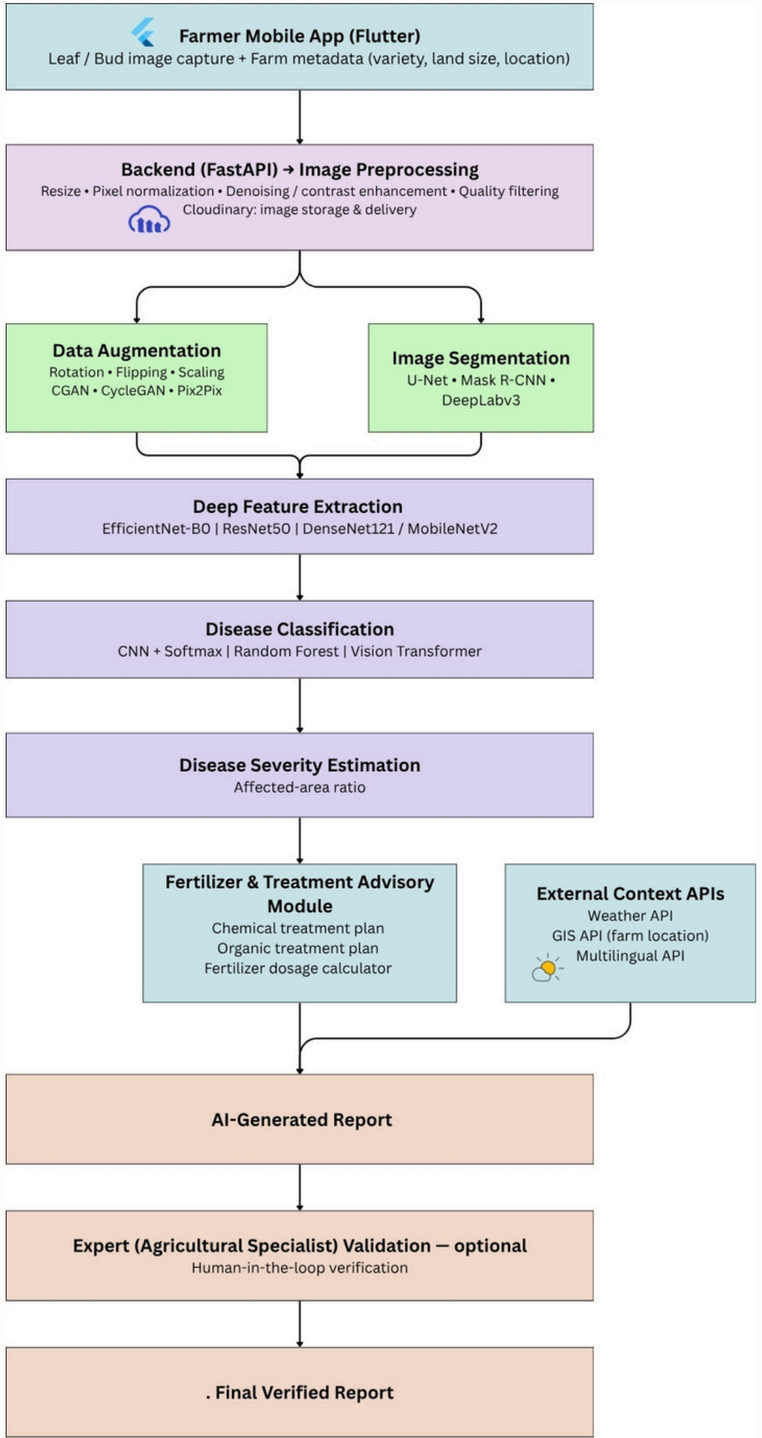
Inputs:
- → Historical yield data
- → Weather data
- → Disease severity

Output: Predicted yield

Advisory System
- AI-generated report
- Treatment suggestions
- Expert validation
- Personalized recommendations

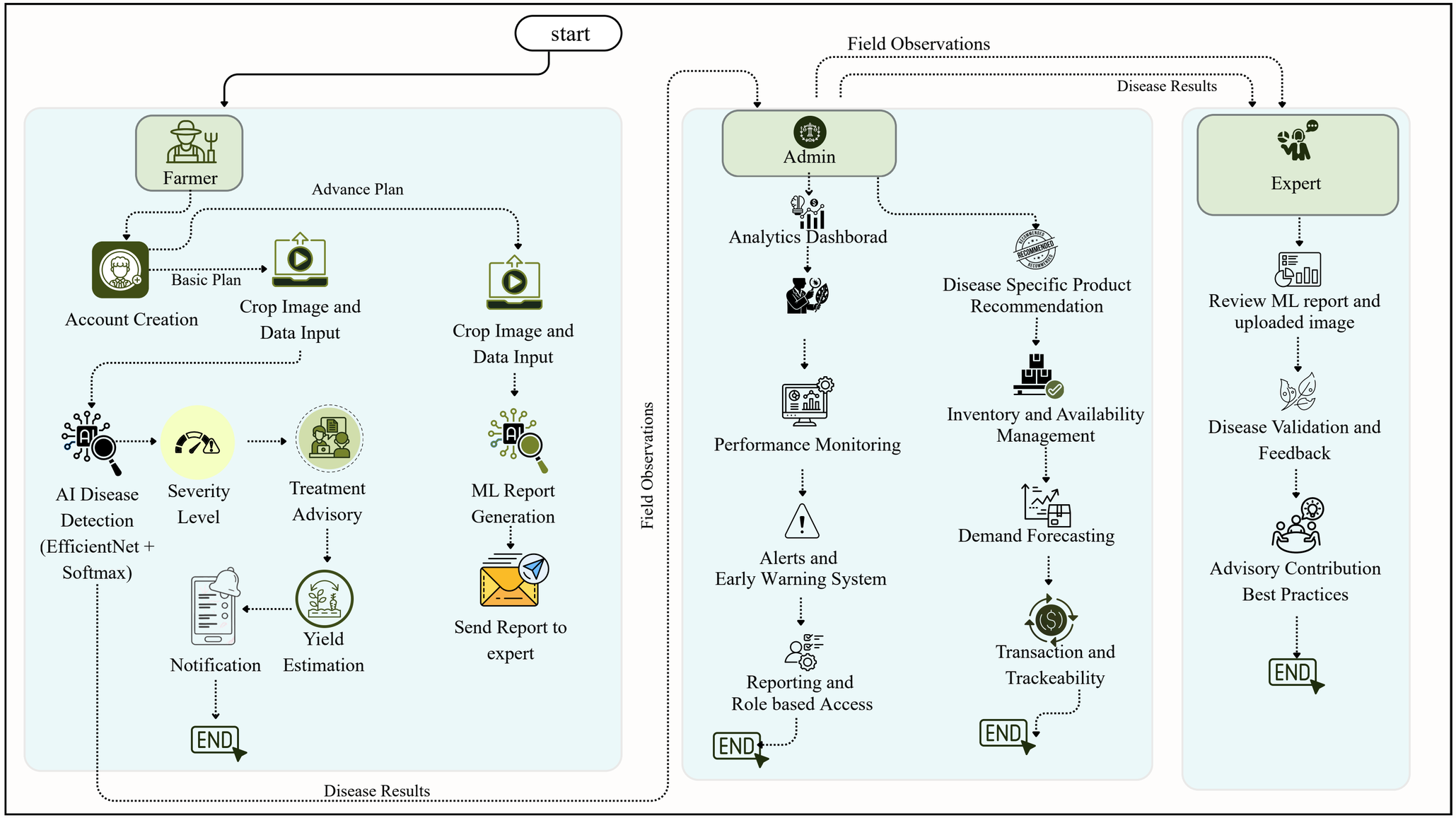
Governance Admin

Monitoring and Management



Methodology Employed

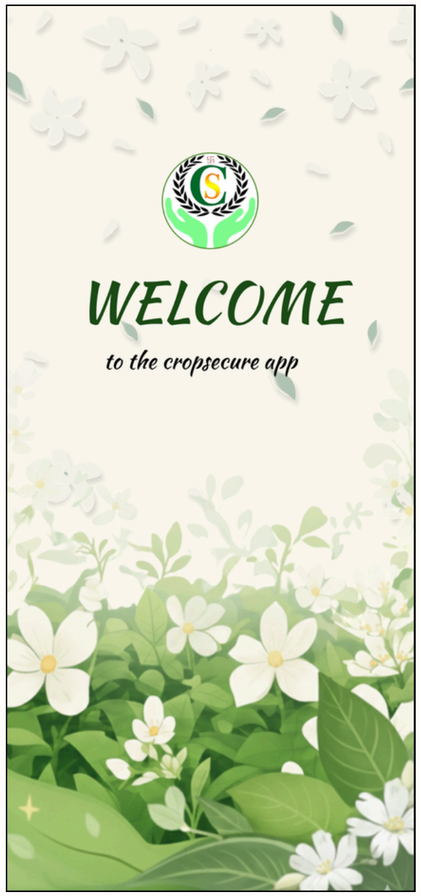
Block Diagram



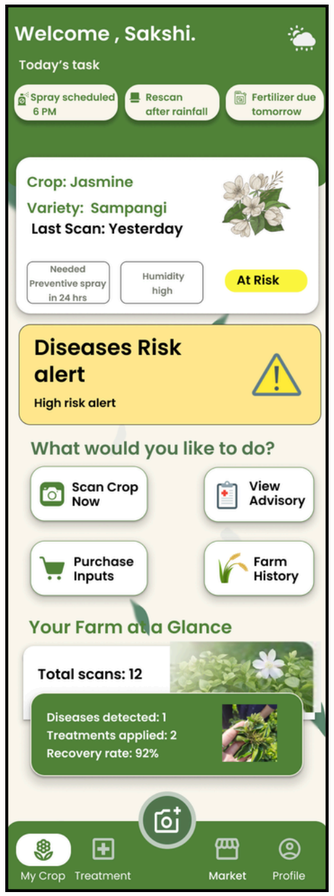
Figma UI

App Figma

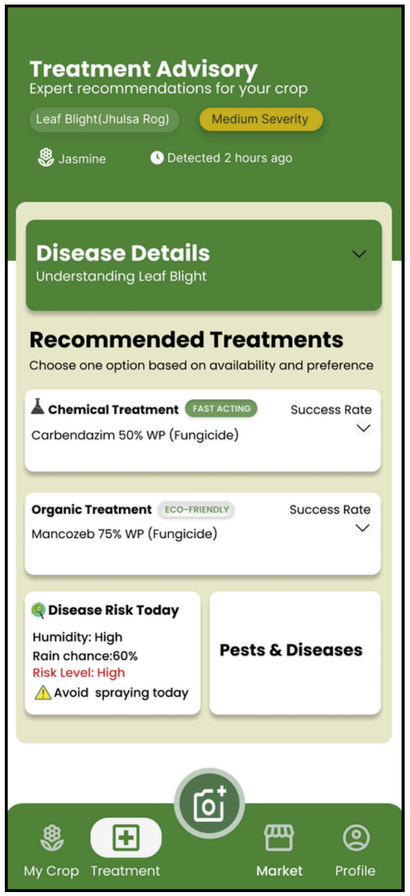
https://www.figma.com/proto/396q0BKpFsAzc0igS0rRbz/CropSecure_MOBILE_App?node-id=0-1&t=Vgwe840yWcPXyfKB-1



This screen serves as the entry point of the CropSecure App



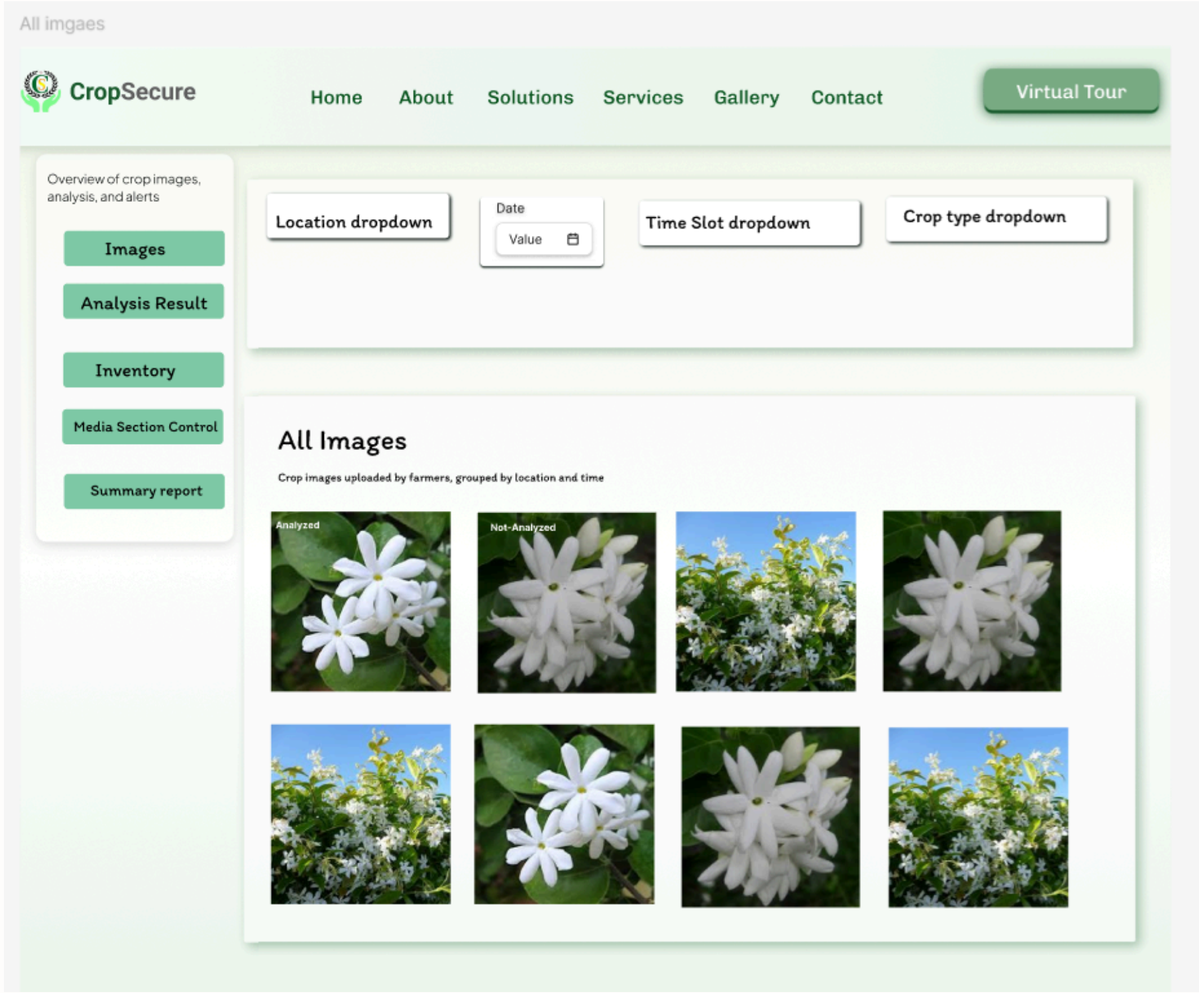
The Main Dashboard displays personalized farm information, including crop type, last scan details, and current risk status.



This screen provides recommended treatments based on the detected disease.

Web Figma

https://www.figma.com/proto/ILTpRBgWJLJN8cAvPIKOI4/CropSecure_WEB_App?node-id=0-1&t=0iX5VvMnamtbFp2H-1



For Full flow refer this PDF https://drive.google.com/file/d/1anJi9vr50P8QH1p1QPlXpFzf3SBdoCmJ/view?usp=drive_link

Implementation (Work Done)

1

Mobile App Design (Flutter)

Designed and developed all screens of the application using Flutter and Dart

2

Mobile App - Database Integration

Connected the mobile application with MongoDB for text and cloudinary for image data storage

3

Dataset Preparation

Collected datasets from research papers and online sources, organized them on GitHub following PyTorch conventions

4

Data Augmentation

Applied augmentation techniques to improve dataset quality and balance class distribution

5

Model Training – Multiple Models

Trained EfficientNet-B0, MobileNet, and ResNet on the prepared dataset and analysed the accuracy

6

Model Selection Process

Evaluated various models and selected EfficientNet-B0 based on highest accuracy

7

Prediction System Implementation

Used the trained EfficientNet-B0 model to predict plant diseases from input images

Conclusion



1. In conclusion, the proposed AI-based smart monitoring system addresses the critical challenges faced in jasmine cultivation by enabling early disease detection, severity assessment, and yield prediction using image analysis and data-driven insights.
2. Unlike existing solutions, this system is specifically designed for jasmine crops and integrates advisory support and real-time monitoring in a single platform.
3. By helping farmers take timely and accurate decisions, the system aims to reduce crop losses, improve productivity, and promote sustainable floriculture practices.

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